# MALL-PE-08: Hardware/Environmental Fault Resolution for Malleswaram Node, Cisco PE Device

**Node:** MALL-PE-08 | **Vendor:** Cisco | **Severity:** Critical | **Alarm Code:** ALARM_1138

## Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve a critical hardware/environmental fault on the Cisco PE device, MALL-PE-08, located at the Malleswaram Node. The failure domain of interest is related to ASIC memory parity errors, line card crashes, and power supply redundancy loss, characterized by the alarm code ALARM_1138. This failure domain occurs due to the inherent design of the Cisco PE device, where the PE role relies heavily on the line cards to process packets and maintain network connectivity. In the event of a line card crash or ASIC memory parity error, the device's ability to forward traffic is compromised, triggering the ALARM_1138. Furthermore, the loss of power supply redundancy can exacerbate the situation, leading to a complete loss of network connectivity. This SOP aims to provide a structured approach to diagnose and resolve this critical failure domain, ensuring minimal downtime and maintaining network reliability.

## Pre-requisite Checks

- Confirm the site location and device ID (MALL-PE-08) prior to initiating troubleshooting.

- Verify the redundant link status to ensure it is operational before proceeding with troubleshooting.

- Dispatch remote hands to the Malleswaram Node to gain physical access to the device, if necessary.

- Ensure all necessary personnel are aware of the potential risks and consequences of troubleshooting a critical failure domain.

## Potential Root Causes

| Cause | Probability |
|---|---|
| ASIC memory parity error | High |
| Line card crash | Medium |
| Power supply redundancy loss | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show platform, Expected Output: Platform information, including line card status and power supply redundancy, Description: This command provides an overview of the device's platform, including the status of the line cards and power supply redundancy. It helps identify potential issues related to the line card crashes or power supply redundancy loss.

Command: show controllers, Expected Output: Controller information, including ASIC memory parity status, Description: This command
```

```
displays information about the device's controllers, including the
ASIC memory parity status. It helps diagnose issues related to ASIC
memory parity errors.

Command: show redundancy, Expected Output: Redundancy information,
including power supply status, Description: This command provides
information about the device's redundancy configuration, including the
power supply status. It helps identify potential issues related to
power supply redundancy loss.
```

## Resolution Steps

Step 1: Step: Replace the failed line card, Command: no boot module , Description: This step involves removing the failed line card from the device and replacing it with a new one. The 'no boot module' command is used to prevent the device from booting up the failed line card.

Step 2: Step: Replace the failed power supply, Command: power redundancy force-activate , Description: This step involves replacing the failed power supply with a new one. The 'power redundancy force-activate' command is used to activate the new power supply and ensure redundancy is restored.

Step 3: Step: Reset the device, Command: reload, Description: This step involves resetting the device to ensure that all configurations and changes are applied.

## Post-Check Verification

```
Command: show platform, Expected Output: Platform information,
including line card status and power supply redundancy, Description:
This command is used to verify that the device has been successfully
reset and that the line cards and power supply redundancy are
functioning correctly.

Command: show redundancy, Expected Output: Redundancy information,
including power supply status, Description: This command is used to
verify that the power supply redundancy has been restored and that the
power supply is functioning correctly.
```

## Roll-back Procedures

- Step: Restore the failed line card, Command: boot module , Description: This step involves restoring the failed line card to its previous state. The 'boot module' command is used to activate the failed line card.

- Step: Restore the failed power supply, Command: power redundancy force-inactive , Description: This step involves restoring the failed power supply to its previous state. The 'power redundancy force-inactive' command is used to deactivate the failed power supply.

- Step: Restart the device, Command: no reload, Description: This step involves restarting the device to ensure that all configurations and changes are restored to their previous state.